

Amit Kulkarni

Personal Details

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Professional Summary

Versatile technology professional with extensive experience in DevOps, cloud architecture, full-stack development, and machine learning. Specializes in implementing end-to-end CI/CD pipelines, cloud infrastructure automation, and developing scalable web and mobile applications. Proven expertise in AWS cloud services and infrastructure as code, complemented by strong capabilities in both MERN/MEAN stack development, native/hybrid mobile development, and machine learning applications.

Educational Details

- Presently pursuing **PhD** from Lovely Professional University in Computer Applications (with research topic as Machine Learning)
- **MCS:** Completed in 2007 with First Class
- **BCS:** Completed in 2003 with First Class
- **Higher Secondary (12th):** Completed in 2000 with First Class
- **Secondary (10th):** Completed in 1998 with First Class

Professional Certifications

- **Kubernetes:**
 - Certified Kubernetes Administrator (CKA)
 - Certified Kubernetes Application Developer (CKAD)
- **Jenkins:** Certified Jenkins Engineer (CJE)
- **Docker:** Certified Docker Associate
- **CompTIA:**
 - Linux+ Certification
 - Security+ Certification

Core Competencies

Full-Stack Development

- **MERN Stack:**
 - MongoDB for NoSQL database management
 - Express.js/NestJS for backend API development
 - React.js for frontend user interfaces
 - Node.js for backend services
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- **MEAN Stack:**
 - MongoDB for database layer
 - Express.js/NestJS for API framework
 - Angular for dynamic frontend applications
 - Node.js for backend services
- **Web Technologies:**
 - RESTful API design and implementation
 - GraphQL API design and implementation
 - State management (Redux, NgRx)
 - Progressive Web Applications (PWA)
 - Responsive design principles

Databases

- **Relational Database:** MySQL, PostgreSQL,
- **NoSQL Databases for Data Persistence:** MongoDB, DynamoDB, Cassandra
- **NoSQL Databases for Caching:** Redis
- **Mobile Database:** SQLite, Realm

DevOps & Infrastructure

- **Container Orchestration:** Advanced expertise in Docker containerization and Kubernetes cluster management.
- **Infrastructure as Code:** Proficient in Terraform for cloud infrastructure provisioning and Ansible for configuration management.
- **CI/CD Implementation:** Experience with Jenkins pipeline creation and ArgoCD for GitOps workflows. Also work with GitHub Actions and GitLab CI.
- **Configuration Management:** Skilled in Ansible and Puppet for automated system configuration and management.
- **Observability:** Implementation of Nagios, Prometheus and Grafana for infrastructure monitoring and alert management.
- **Version Control:** Advanced Git workflows and branching strategies.
- **Virtual Development:** Vagrant for creating and managing development environments.

Cloud Architecture (AWS)

- **Compute Services:** EC2, ECS, EKS, Lambda, Auto Scaling
- **Security & Identity:** IAM roles, policies, and security group management
- **Database Services:** RDS, DynamoDB, Aurora, ElastiCache
- **Networking:** VPC, Route53, CloudFront, API Gateway
- **Storage Solutions:** S3, EBS, EFS
- **Monitoring & Logging:** CloudWatch, CloudTrail
- **DevOps Services:** CodePipeline, CodeBuild, CodeDeploy, CodeCommit
- **Integration Services:** SQS, SES, SNS

Mobile Development

- **Native Development:**
 - Android development using Java and Kotlin
 - iOS development using Swift
- **Cross-Platform Development:**
 - React Native for hybrid application development
 - Experience in bridging native and JavaScript functionalities

Machine Learning & AI

- **Frameworks:** TensorFlow and PyTorch expertise
- **GenAI Applications:** Development of generative AI solutions
- **Python:** Advanced programming for ML/AI applications
- **Model Deployment:** Experience in deploying ML models in production environments

Key Achievements

- Designed and implemented fully automated CI/CD pipelines reducing deployment time by 70%
- Developed scalable web applications using MERN/MEAN stack serving 100K+ daily active users
- Successfully migrated legacy applications to containerized microservices architecture
- Automated infrastructure provisioning reducing setup time from days to hours
- Implemented cost-optimization strategies in AWS, achieving 40% reduction in cloud expenses
- Developed and deployed multiple cross-platform mobile applications using React Native
- Created ML models for production use cases with successful integration into existing systems

Technical Approach

- Advocates for Infrastructure as Code and GitOps methodologies
- Implements security best practices across the development lifecycle
- Focuses on scalable and maintainable architecture design

- Emphasizes monitoring and observability in all deployments
- Promotes automation and self-healing systems
- Implements cost-effective cloud solutions without compromising performance
- Developed more than 90 mobile applications in native and hybrid stack, among which few were already listed in top 5 in both the markets
- Developed many Angular, React and PHP websites for various clients

Areas of Expertise

- Full-Stack Web Development
 - MERN/MEAN stack applications
 - RESTful and GraphQL API design
 - Frontend application architecture
 - Designing Micro-frontend solutions
 - Database optimization
 - Performance tuning
 - Scalable web solutions
 - Designing scalable and highly available microservices backend
- DevOps Pipeline Implementation
 - Source control management
 - Automated testing and quality assurance
 - Continuous integration and deployment
 - Infrastructure automation
 - Configuration management
 - Monitoring and logging
- Cloud Architecture
 - Microservices architecture
 - Serverless computing
 - High availability design
 - Disaster recovery planning
 - Security implementation
 - Cost optimization
- Application Development
 - Native mobile applications
 - Cross-platform solutions
 - API development and integration
 - Machine learning model deployment
 - System architecture design
 - Performance optimization